

TEJA GURUVELLI

US Citizen | gguruvelli3@gatech.edu | linkedin.com/in/tjguruvelli | github.com/tj-guruvelli.com | Willing to Relocate

SUMMARY

Cybersecurity and machine learning specialist with expertise in AI/ML, anomaly detection, and security analytics. Demonstrated success implementing security solutions across automotive, financial, and energy sectors. Passionate about leveraging innovative technologies to solve complex security challenges, with proven ability to adapt quickly to new domains while delivering robust technical solutions

RELEVANT EXPERIENCE

Automotive Security Research Intern Lemont IL Argonne National Laboratory	May 2024 to Aug 2024
- Built real-time threat detection system for connected vehicles using Python, TensorFlow, PyTorch, and CUDA acceleration, achieving 98% detection accuracy while processing vehicle safety messages in under 0.11 milliseconds - Reduced false security alerts by 20% by designing two-phase detection framework combining lightweight statistical analysis with deep learning models, enabling faster threat response for autonomous vehicle networks - Validated system performance through 108 hours of simulation testing across 7 attack scenarios with 49,000+ vehicles, ensuring reliable security in real-world traffic conditions - Demonstrated deployment feasibility by implementing hardware-in-the-loop testing on Raspberry Pi devices, proving system could run on resource-constrained vehicle computers while maintaining real-time threat detection	
Information Security Engineer Intern Des Moines IA Principal Financial Group	May 2023 to Aug 2023
- Engineered 3 automated insider risk detection alerts using Elasticsearch and AWS Lambda, reducing false positives by 30% and eliminating manual daily reviews for SOC analysts - Created security analytics dashboard integrating 4 data sources (Proofpoint, Zscaler) with AWS cloud technologies (S3, Lambda, CloudWatch, SNS), processing log data to detect policy violations - Designed multi-phase alert escalation system with 95% availability, developing playbooks and SIEM integration to automate incident response workflows and SOAR automation for SaaS security - Led software testing and analysis efforts to identify and resolve system deficiencies, supporting more reliable and secure threat detection in financial applications	
Manufacturing Test Engineer Co-op West Union SC Itron	Aug 2022 to May 2023
- Consolidated 6 web-based PHP/MySQL tools into a single C# Windows desktop application with CRUD operations, improving data integrity and system efficiency for quality control teams generating manufacturing reports for Duke Energy and Dominion Energy meter production - Optimized database performance by migrating from MySQL to MS SQL Server, implementing indexes, constraints, and optimized data structures to reduce query retrieval time and enable scalable architecture for manufacturing operations - Improved Overall Equipment Effectiveness (OEE) by developing automated validation protocols that reduced user errors and strengthened quality control processes, supported by comprehensive technical documentation using version control for collaborative development - Reduced new team member onboarding time by authoring detailed technical documentation for manufacturing test procedures, ensuring knowledge transfer and supporting ongoing software maintenance across Windows-based production systems	
Software Engineer Intern Clemson SC Clemson Energy, Visualization & Analytics Center (CEVAC)	May 2022 - Aug 2022
- Developed automated Python data pipeline processing 100M+ records using optimized data structures and algorithms, eliminating manual interventions and deploying on Linux servers to provide scalable facility monitoring solutions - Developed full-stack web application using ASP.NET, C#, JavaScript, and MS SQL for administrative console managing real-time sensor data across 32 buildings, tracking CO2, humidity, temperature, and energy metrics for environmental monitoring - Migrated scheduler system and backend infrastructure to new server architecture, transitioning from Cron-based scheduling to custom Python solution that automated data collection workflows and improved ETL processing efficiency - Optimized data infrastructure by integrating scalable design solutions to support high-volume processing, ensuring seamless system reliability	

EDUCATION

M.S. Cybersecurity Georgia Institute of Technology Atlanta, GA	Expected in May 2027
B.S. Computer Science, Minor in Cybersecurity Clemson University Clemson, SC	Dec 2024

PROJECTS

NVIDIA Capstone Project Clemson, SC Clemson University School of Computing Capstone Program (Product Developer)	Aug 2024 – Dec 2024
- Benchmarked 9 large language models using CUDA-accelerated NVIDIA GPUs, identifying optimal models for educational applications and reducing inference latency through GPU optimization - Improved model accuracy by 11% (from 64% to 71.5% on MMLU benchmark) by implementing Parameter-Efficient Fine-Tuning (PEFT) on 8,000 synthetically generated questions, utilizing 4,000+ A100 GPU hours - Built RAG-based virtual teaching assistant integrating NeMo Guardrails with Canvas LMS and FAISS vector database, deploying across AWS cloud infrastructure (EC2, Docker) to serve 1,000+ students	
Meaning Alignment Institute Clemson, SC AI Safety & Ethics Developer	Aug 2024 – Dec 2024
- Designed an evaluation metrics system using JSON and Python (Pandas, NumPy, Scikit-learn) to analyze 4,500+ AI interaction logs, enhancing ethical reasoning depth by 20-25% through expanded frameworks - Built a machine learning pipeline for ethical prompt engineering, improving contradiction handling by 10-15% with iterative refinements and feedback loop mechanisms - Evaluated AI model outputs across 4 ethical frameworks, utilitarianism, deontology, virtue ethics, and social contract theory, boosting flawed reasoning detection by 18%	

SKILLS

Languages: Python, C#, Java, C/C++, JavaScript, SQL, React.js, CSS, ASP.NET, .NET CORE ML/AI Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, NVIDIA NeMo, XGBoost, Pandas, NumPy, SciPy Cloud & Infrastructure: AWS (EC2, S3, Lambda, CloudWatch, SNS), Azure (Data Factory, Fabric), Linux, Windows, Docker, Kubernetes, CUDA Data Engineering: MS SQL Server, MySQL, PostgreSQL, Snowflake, Elasticsearch, FAISS, Splunk, Power BI, ETL pipelines Tools & Technologies: Git, Wireshark, Jira, Confluence, Postman, Swagger Specializations: Deep learning, anomaly detection, data structures & algorithms, RAG systems, PEFT, version control, Agile methodologies	
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